

Predictive Analytics Using Historical Data

Revenue Forecasting Using Linear Regression

Internship Project – Task 3

Submitted by:

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Tools Used:

Python | Google Colab | Pandas | Scikit-learn | Matplotlib

1. Introduction

Predictive analytics uses historical data, statistical techniques, and machine learning models to identify patterns and predict future outcomes. In this project, historical monthly revenue data is analyzed to understand the revenue trend and forecast future revenue.

The project demonstrates how predictive modeling can help businesses make better decisions based on historical performance.

2. Objective

The main objectives of this project are:

- To clean and preprocess historical revenue data.
- To analyze historical revenue trends.
- To build a predictive model using Linear Regression.
- To evaluate the model using MAE, RMSE, and R^2 Score.
- To visualize actual and predicted revenue.
- To forecast revenue for the next six months.

3. Dataset

The dataset used in this project contains historical monthly revenue data from January 2025 to August 2026. It consists of two main variables:

- **Month** – Represents the month of the recorded revenue.
- **Revenue** – Represents the monthly revenue value.

The dataset contains **20 monthly observations** and was created as a sample historical dataset for demonstrating predictive analytics.

Dataset Preview

Now go back to **Google Colab** and find the table where you have:

	Month	Revenue	Month_Number
0	2025-01-01	120000	1
1	2025-02-01	125000	2
2	2025-03-01	132000	3
3	2025-04-01	128000	4
4	2025-05-01	140000	5

4. Data Preprocessing

The historical data was checked and prepared before applying the predictive model. The following preprocessing steps were performed:

1. Checked the dataset structure and data types.
2. Checked for missing values.
3. Converted the Month column into a proper date format.
4. Created a numerical **Month Number** variable to represent the chronological order of the data.
5. Divided the dataset into training and testing data using an 80:20 split.

The chronological order of the data was maintained during the train-test split because the project involves historical time-based data.

5. Methodology

The following methodology was used to build the predictive analytics model:

5.1 Data Preparation

The historical monthly revenue data was cleaned and converted into a suitable format for analysis.

5.2 Feature Creation

A numerical **Month Number** was created to represent the chronological order of the months. This variable was used as the input feature for the prediction model.

5.3 Train-Test Split

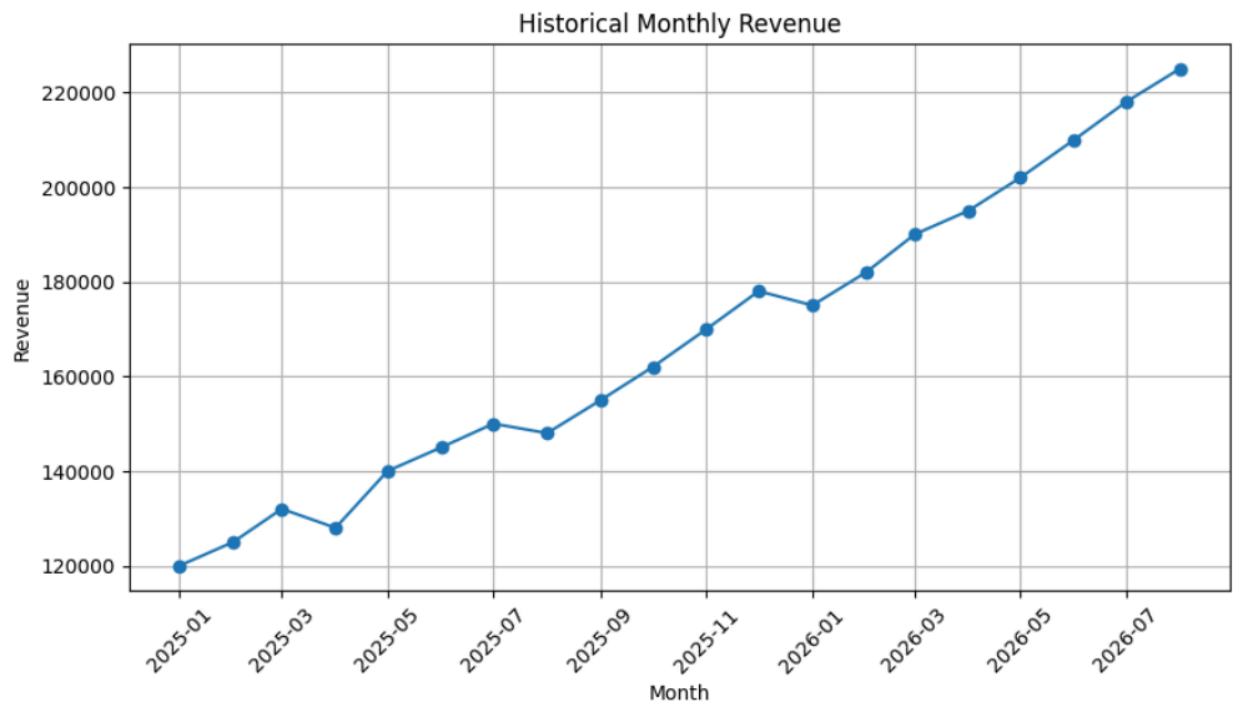
The dataset was divided into:

- **80% training data** – used to train the model.
- **20% testing data** – used to evaluate the model.

The data was not shuffled so that the chronological order of the historical data was maintained.

5.4 Predictive Model

Linear Regression was selected as the predictive model. The model learns the relationship between the month number and historical revenue and uses this relationship to estimate future revenue.

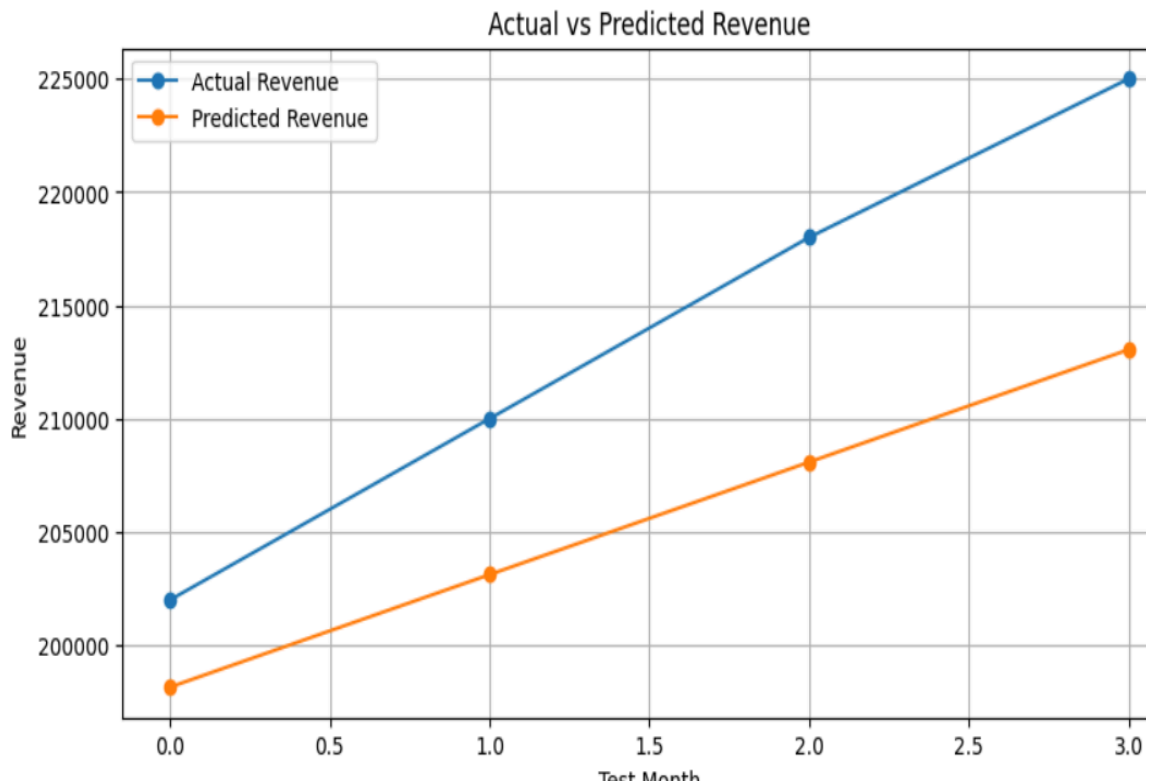


6. Model Training and Prediction

A Linear Regression model was used to predict revenue based on the chronological month number. The dataset was divided into training and testing sets using an 80:20 split. The training data was used to teach the model the relationship between time and revenue, while the testing data was used to evaluate its prediction performance.

Actual vs Predicted Revenue

The actual and predicted revenue values were compared using a line graph. This visualization helps to understand how closely the model's predictions match the actual historical revenue values.



7. Model Evaluation

The performance of the Linear Regression model was evaluated using three metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. These metrics help measure the accuracy and performance of the predictive model.

MAE: **8150.74**

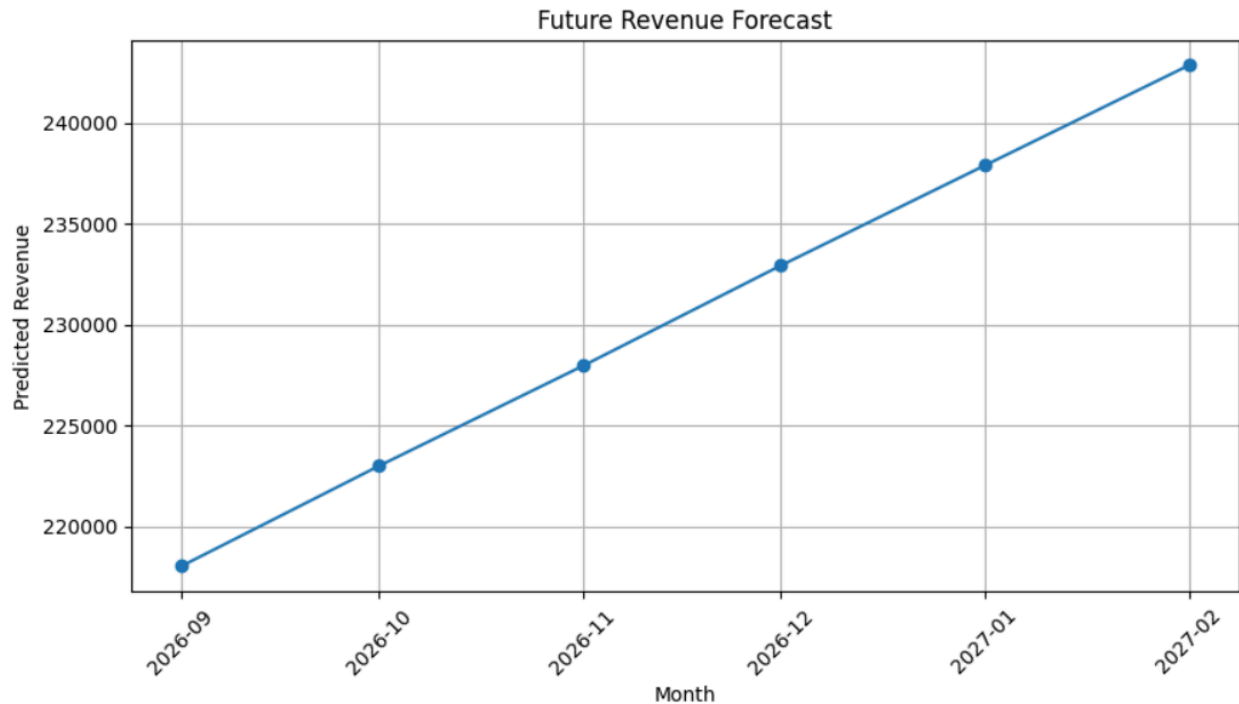
RMSE: **8709.29**

R^2 Score: **-0.0224**

8. Future Revenue Forecast

After evaluating the model, the trained Linear Regression model was used to forecast future revenue for the next six months based on the historical revenue trend.

index	Month	Predicted_Revenue
0	2026-09-01 00:00:00	218014.71
1	2026-10-01 00:00:00	222980.88
2	2026-11-01 00:00:00	227947.06
3	2026-12-01 00:00:00	232913.24
4	2027-01-01 00:00:00	237879.41
5	2027-02-01 00:00:00	242845.59



The forecast results provide an estimate of the expected revenue for the next six months based on the historical revenue pattern. These predictions can help in understanding the possible future revenue trend and support basic business planning and decision-making.

9. Key Findings

- Historical revenue shows an overall increasing trend with some month-to-month fluctuations.
- The Linear Regression model was trained using historical monthly revenue data.
- The model's performance was evaluated using MAE, RMSE, and R^2 Score.
- The model was used to forecast revenue for the next six months.
- The forecast provides an estimated view of future revenue based on the historical trend.

10. Conclusion

This project demonstrated how historical revenue data can be used for predictive analytics and future forecasting. A Linear Regression model was developed using historical monthly revenue data and evaluated using standard performance metrics. The results and visualizations helped analyze the relationship between historical and predicted revenue. The model was also used to generate a six-month future revenue forecast, demonstrating how predictive analytics can support trend analysis and business planning.